

AgentBoard: An Analytical Evaluation Board of Multi-turn LLM Agents

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AgentBoard: Title and Authors

Benchmark and toolkit for multi-turn LLM agents with fine-grained progress diagnostics.

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Motivation Beyond Success

- Success-only metrics mask differences in challenging multi-round settings.
- Evaluation needs diverse tasks, partial observability, and deep interaction.
- Progress rate reveals improvements when success is near zero.
- Generalist agents require **grounding** and **world modeling** assessment.
- Prior benchmarks emphasize final scores over [incremental progress](#).

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Why AgentBoard

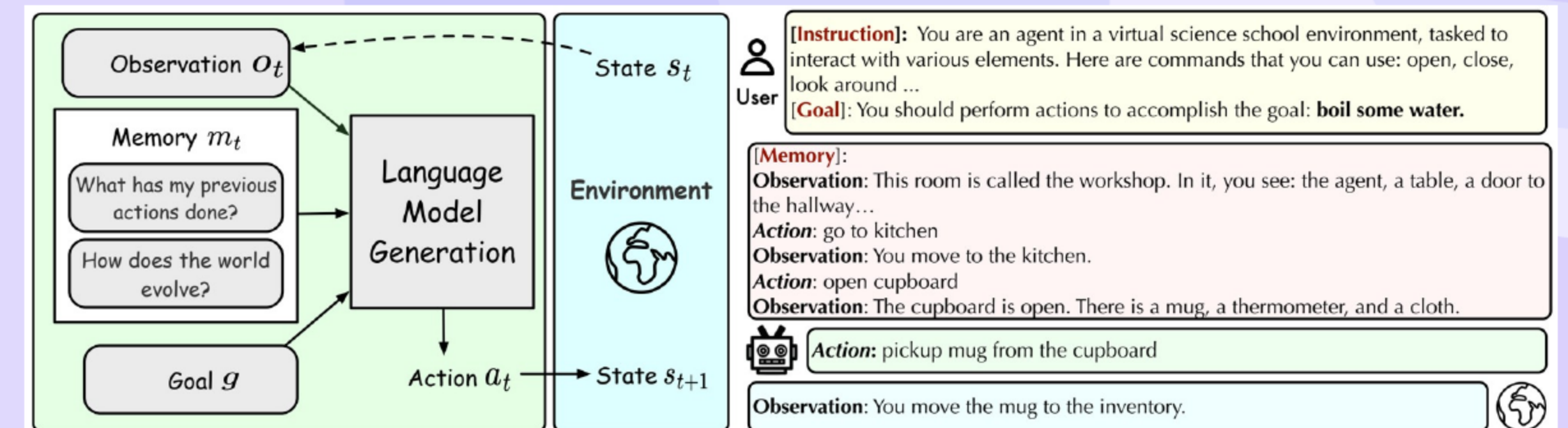
- Nine task types and 1013 environments across text-only scenarios.
- Subgoals enable a unified **progress rate** beyond final success.
- Analytical panel supports fine-grained **diagnosis** and drill-down.
- All environments are multi-round and partially observable.
- Design isolates decision-making without multimodal confounds.

- Tasks and analytical evaluation framework at a glance.

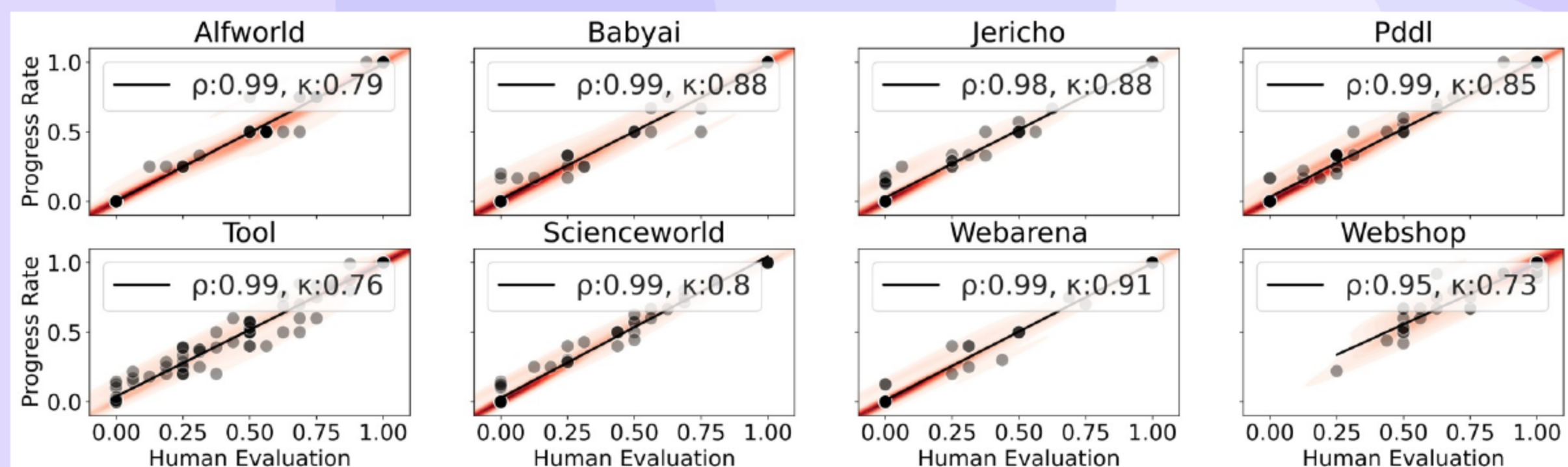
- Unified protocol across nine environments reports success and progress.
- GPT-4 [OpenAI '23] leads overall; progress separates models when success is near zero.
- Scaling and instruction tuning help, yet open-weight gaps remain.
- Stronger agents sustain **long-horizon** interactions and planning.
- Diagnostics expose **action-format errors** and grounding issues.
- Baselines: GPT-4 [OpenAI '23], Claude [Anthropic '23], Gemini [Google '23], Llama 2 [Touvron '23], Llama 3 [Meta '24], Mistral 7B [Mistral '23], Code Llama [Rozière '23]

OpenAI '23; Anthropic '23; Google '23; Touvron '23; Meta '24; Mistral '23; Rozière '23; Chiang '23; DeepSeek '24; Xu '24; Wang '24; Mistral 7B [Mistral '23]; Code Llama [Rozière '23]

- Unified benchmark with diverse, multi-round, partially observable tasks.
- Fine-grained **progress rate** for sensitive improvement tracking.
- Comprehensive analyses of trajectories, grounding, and sub-skills.
- Interactive panel enables **efficient diagnosis** and reporting.
- Task design correlates with **reasoning** and **coding** strengths.



- Reflex agent uses act-only prompting with sliding-window memory.
- Environment feedback lists valid actions, executes changes, and returns errors.
- Design supports long interactions without exceeding context limits.
- Framework is compatible with **ReAct** and other prompting variants.
- Error handling improves **robustness** across domains.



- Metric captures partial completion across multi-round interactions.
- Direct state matching applies when goals are comparable to states.
- Ordered subgoals use labeled checks with simple pattern matching.
- Reporting includes **both** success and progress for sensitivity.
- Design highlights **steady advances** obscured by zero success.
- Dots=instances; x=human score, y=progress rate; ρ =correlation, κ =agreement.

- AlfWorld shows correct navigation and container manipulation before misidentification stalls completion.
- Jericho demonstrates discovery and unlocking behaviors before failing a distraction step.
- WebArena completes authentication and form entries while missing a required control for submission.
- Progress traces reveal **incremental gains** despite zero final success.
- Trajectories expose **plateaus** linked to grounding and workflow errors.

- Goal: escape the cell; discovers a hidden key in the bunk.
- Unlocks the door and steps into the corridor.
- Fails to distract the guard; forced back to the cell.
- Shows grounded world-model updates and partial plan execution.

- Goal: file a support ticket on a live site.
- Reaches site, logs in, navigates to support page.
- Fills most fields correctly; misses a required dropdown.
- Progress rises steadily despite zero final success.

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Curation and Reporting

- Human-labeled subgoals are validated to ensure unique, sensible sequences.
- Reporting pairs success with progress for improved sensitivity.
- Inter-annotator agreement supports [reliable](#) trajectory scoring.
- Light edits improve clarity without reducing trajectory diversity.
- Checks prioritize **consistency** over exhaustive coverage.

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Embodied and Games

- Embodied tasks include AlfWorld, ScienceWorld, and BabyAI with textual interfaces.
- Games include Jericho and PDDL domains with symbolic constraints.
- Tasks emphasize **planning**, **navigation**, and partial observability.
- Progress metrics align with domain specifics for fair evaluation.
- Design choices target [generalization](#) beyond single domains.

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Web and Tools

- Web tasks include WebShop and WebArena with realistic workflows.
- Tool tasks include Tool-Query and Tool-Operation with structured APIs.
- Evaluation covers **authentication**, **form filling**, and parameter grounding.
- Feedback improves handling of [invalid](#) and out-of-scope actions.
- Progress uses domain-aligned matching for [sensitive](#) scoring.

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Backup: What Improves Agents

- Coding-heavy pretraining and instruction tuning improve progress across domains.
- GPT-4 [OpenAI '23] leads overall, while open-weight models vary by task strength.
- Easy versus hard splits show consistent drops across all models.
- Grounding accuracy correlates with [robust](#) long-horizon performance.
- Sub-skill analysis highlights planning and self-reflection as key drivers.

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Panel Case Study

- Weights and Biases panel enables per-task drill-down and trajectory inspection.
- Diagnostics identify plateaus, grounding issues, and workflow failures.
- Case studies reveal **formula semantics** gaps and parameter errors.
- Comparisons separate grounding from **plan repair** weaknesses (e.g., GPT-4 [OpenAI '23], Llama 3 [Meta '24], Mistral 7B [Mistral '23], Claude [Anthropic '23]).
- Summary and task boards streamline **actionable** analysis.

OpenAI 2023; Anthropic 2023; Meta 2024; Mistral 2023; DeepSeek 2024; Chiang 2023

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Analytical Panel Figure

- Dashboard view supporting drill-down and trajectory inspection.
- Panes: overall summary, per-task boards, and trajectory views.

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Conclusion and Limitations

- AgentBoard offers a unified benchmark, progress metric, and analytical toolkit.
- Human-annotated subgoals introduce **annotation bias** and maintenance cost.
- Simulated settings limit realism despite **standardized** evaluation.
- Future work targets real-world tasks and automated subgoal generation.
- Toolkit aims to support **reproducible** and **scalable** agent evaluation.

Thank You

Questions & Discussion